

Robotics Kinematics

BRAC University

Forward Kinematics

Method 1: Using Trigonometric Process (Previous Class)

Limitations:

- Becomes Complicated for Multiple Links
- Not Systematic (Depends on geometry insight, Requires manual thinking every time, No fixed procedure)
- Hard to Generalize (Change joint orientation: re-derive everything again)
- etc

Forward Kinematics

Method 2: Using Homogeneous Transformations (Recall Linear Algebra Course)

Process:

1. Define the coordinates of frames
2. Finding Denavit-Hartenberg Parameters, *Northwestern Engineering's legacy in robotics started in the 1950s when Dick Hartenberg, a professor, and Jacques Denavit, a PhD student, developed a way to represent mathematically how mechanisms move* <https://robotics.northwestern.edu/history.html>
3. Building Homogeneous Transformation Matrix for the Robot

D-H Parameter Method

Definition:

A standardized framework to describe relative positions and orientations of adjacent links in a robot

Benefit:

Provides a systematic method to describe the geometry of a robot, making it easier to perform forward and inverse kinematics calculations.

Forward Kinematics

Before moving to Homogeneous Transformations Process, we should know about first



D-H Parameter Calculation



Links are required to figure out the joints. All calculations are done using the axes of the joint.



For each joint, we have **4** parameters.

Definition of the Four Parameters

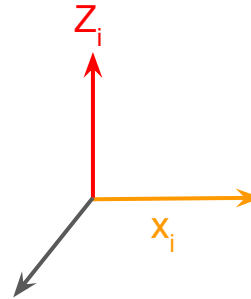
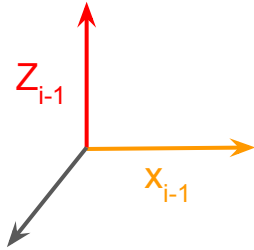
1. Link Length (a_i): Distance between z_{i-1} and z_i along x_i
2. Link Twist (α_i): Angle between z_{i-1} and z_i about x_i
3. Joint Offset (d_i): Distance between x_{i-1} and x_i along z_{i-1}
4. Joint Angle (θ_i): Angle between x_{i-1} and x_i about z_{i-1}

Note: Don't try to imagine the Y-axis

<https://www.youtube.com/watch?v=rA9tm0gTln8>

Link Length (a_i)

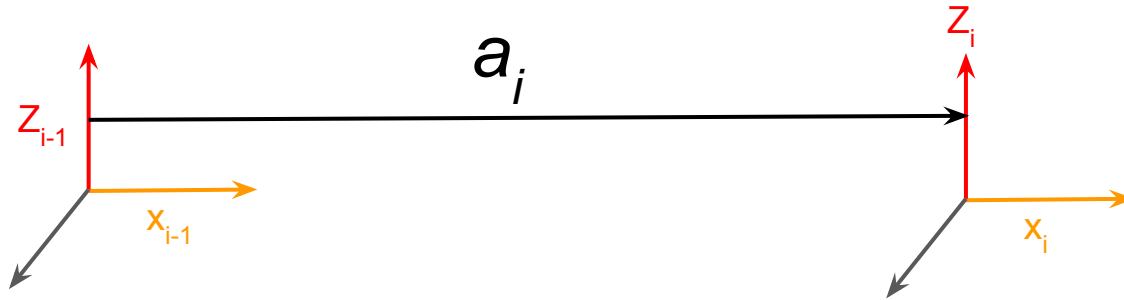
Distance between z_{i-1} and z_i along x_i



If you start walking, starting from z_{i-1} to z_p using the road x_i . What will be the distance?

Link Length (a_i)

Distance between z_{i-1} and z_i along x_i

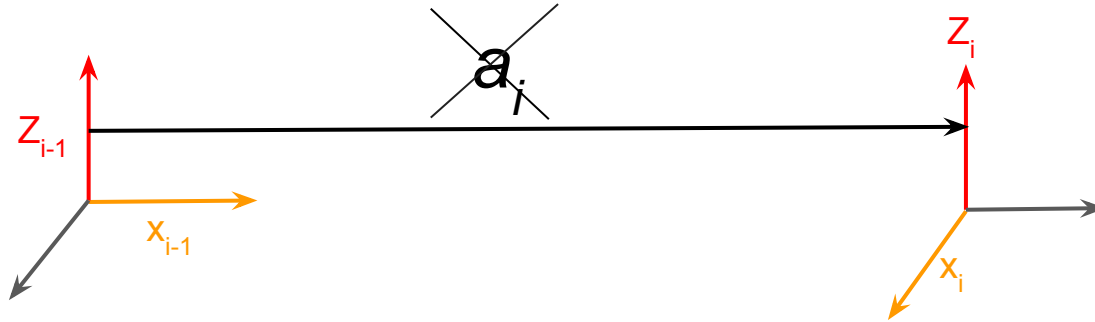


The black line is that distance. Very Easy Right !!!

Now let's make the second frame rotate.

Link Length (a_i)

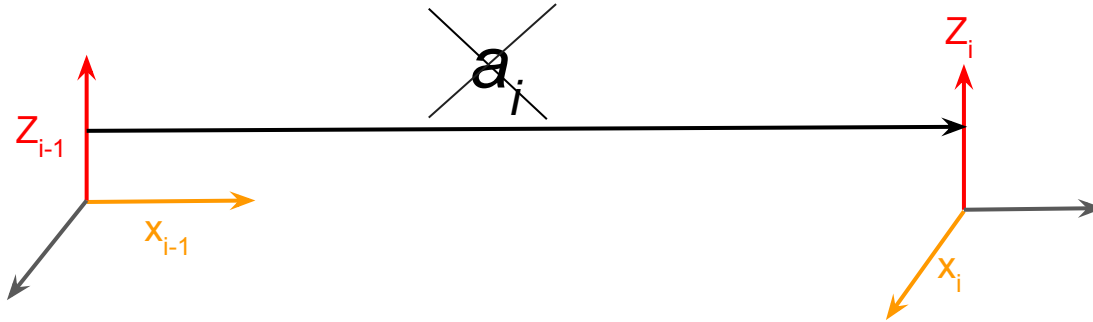
Distance between z_{i-1} and z_i along x_i



Now a_i is not the link length, why??

Link Length (a_i)

Distance between z_{i-1} and z_i along x_i



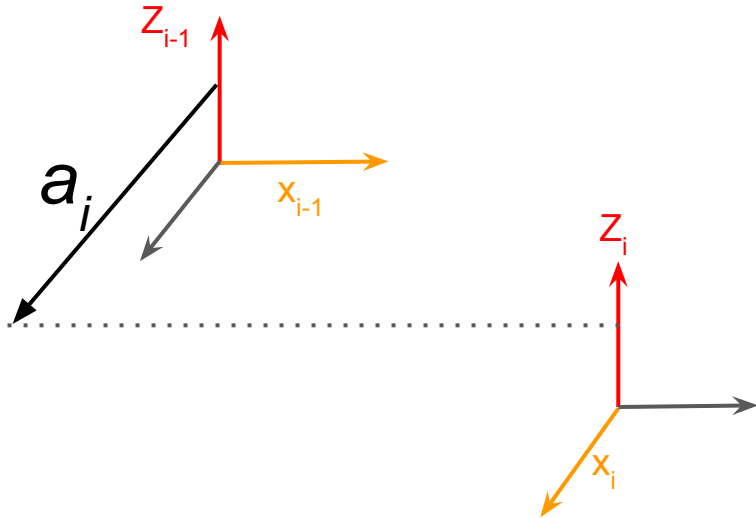
Here the you are not walking along x_i axis to reach z_i instead y_i axis.

In this example, there is no distance between z axis along x_i axis.

$$a_i = 0$$

Link Length (a_i)

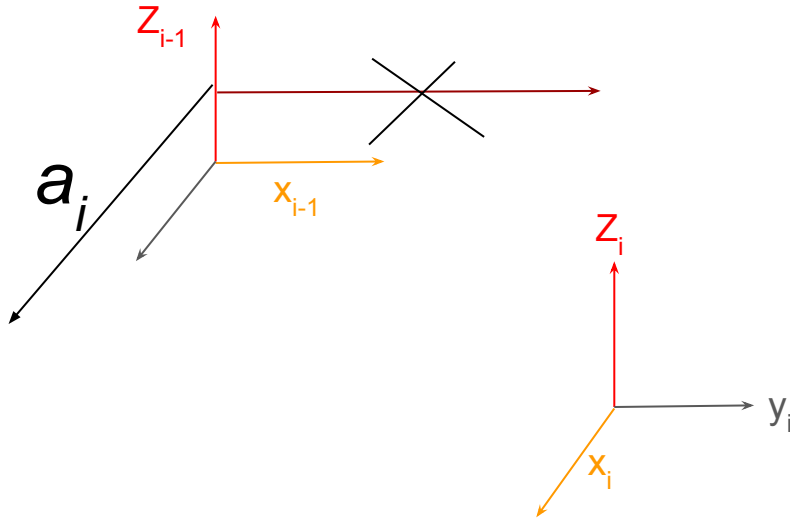
Distance between z_{i-1} and z_i along x_i



Here, you can see the a_i distance is along x_i axis.

Link Length (a_i)

Distance between z_{i-1} and z_i along x_i

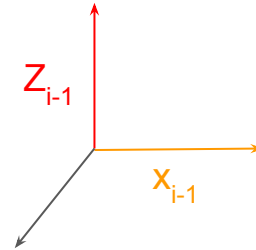
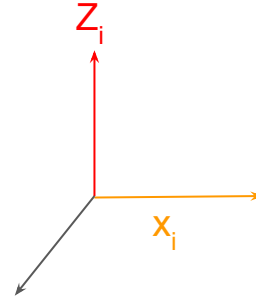


Confusion: You may get confused about the **Red** line distance. This line actually the moving along y_i axis. That's why according to the the definition that will not the Link length.

Joint Offset (d_i)

Distance between x_{i-1} and x_i along z_{i-1}

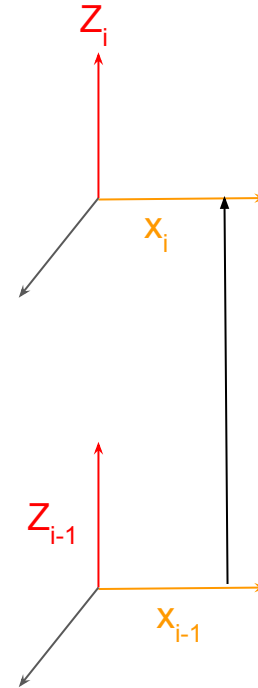
If you start walking, starting from x_{i-1} to x_i using the road z_{i-1} What will be the distance?



Joint Offset (d_i)

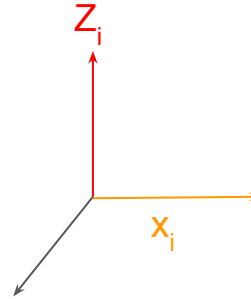
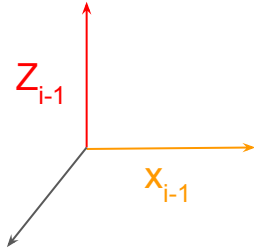
Distance between x_{i-1} and x_i along z_{i-1}

The black line is that distance. As the distance start from x_{i-1} to x_i along in the direction of z_{i-1}



Joint Offset (d_i)

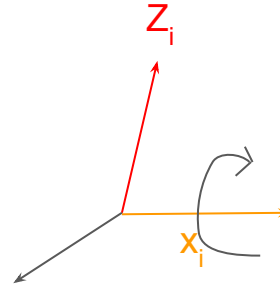
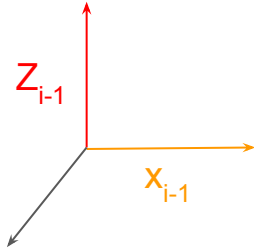
Distance between x_{i-1} and x_i along z_{i-1}



Here $d_i = 0$. Why???

Link Twist (α_i):

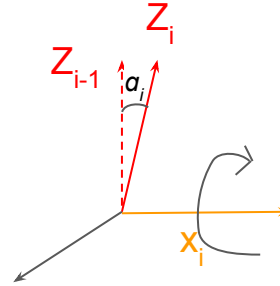
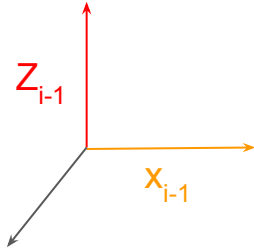
Angle between z_{i-1} and z_i about x_i



When you rotate the the second frame about x_i , there is an angle between z_{i-1} and z_i .

Link Twist (α_i):

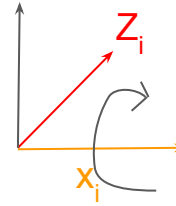
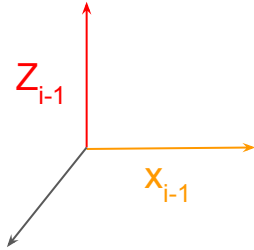
Angle between z_{i-1} and z_i about x_i



When you rotate the the second frame about x_i , there is an angle between z_{i-1} and z_i .

Link Twist (α_i):

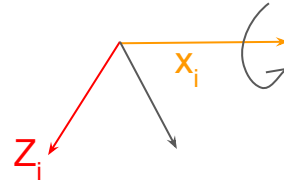
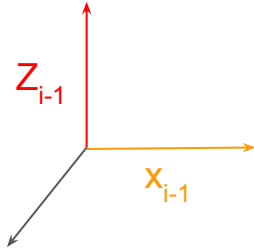
Angle between $\mathbf{z}_{i-1}$ and $\mathbf{z}_i$ about $\mathbf{x}_i$



Here the $\alpha_i = +90^\circ$.

Link Twist (α_i):

Angle between z_{i-1} and z_i about x_i

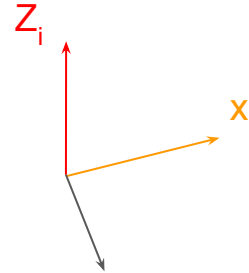
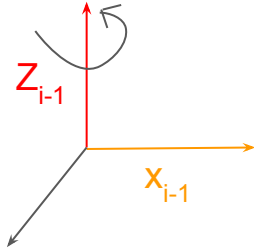


Here the $\alpha_i = -90^\circ$.

Why???

Joint Angle (Θ_i)

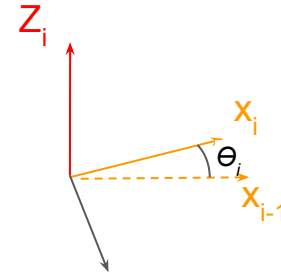
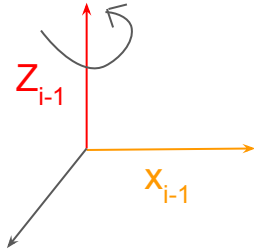
Angle between x_{i-1} and x_i about z_{i-1}



When you rotate the the second frame about z_{i-1} , there is an angle between x_{i-1} and x_i .

Joint Angle (Θ_i)

Angle between x_{i-1} and x_i about z_{i-1}

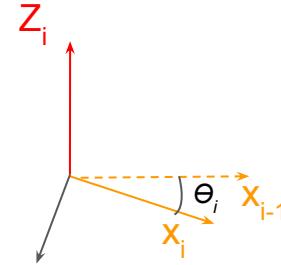
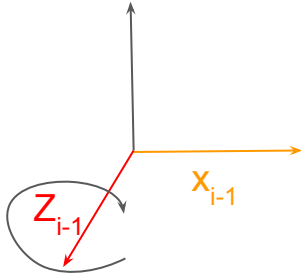


When you rotate the the second frame about z_{i-1} , there is an angle between x_{i-1} and x_i .

Here the Θ_i is the joint angle and it is positive (CCW)

Joint Angle (Θ_i)

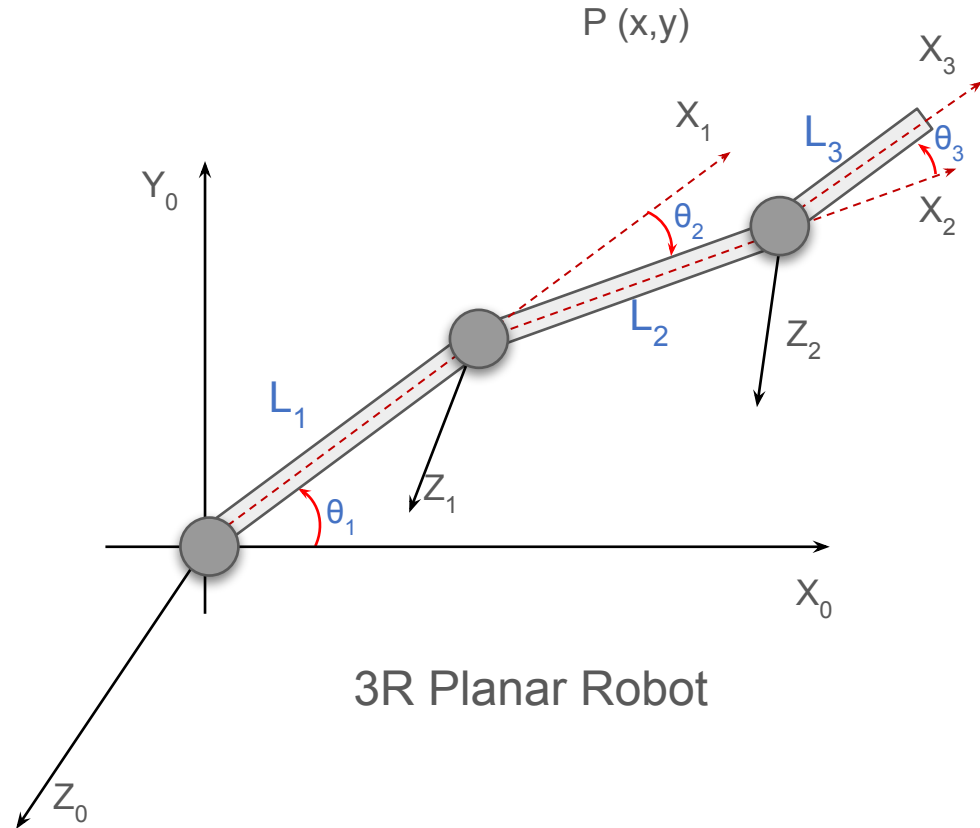
Angle between x_{i-1} and x_i about z_{i-1}



Now Θ_i is negative. Why???

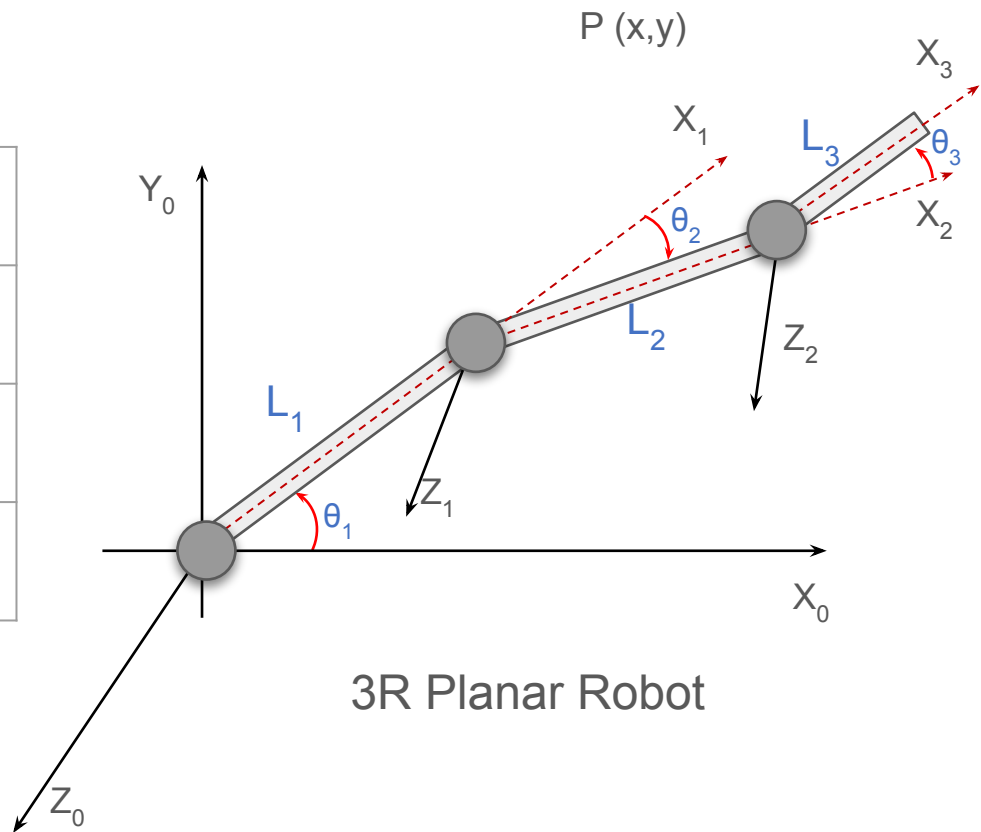
Example

DH Table



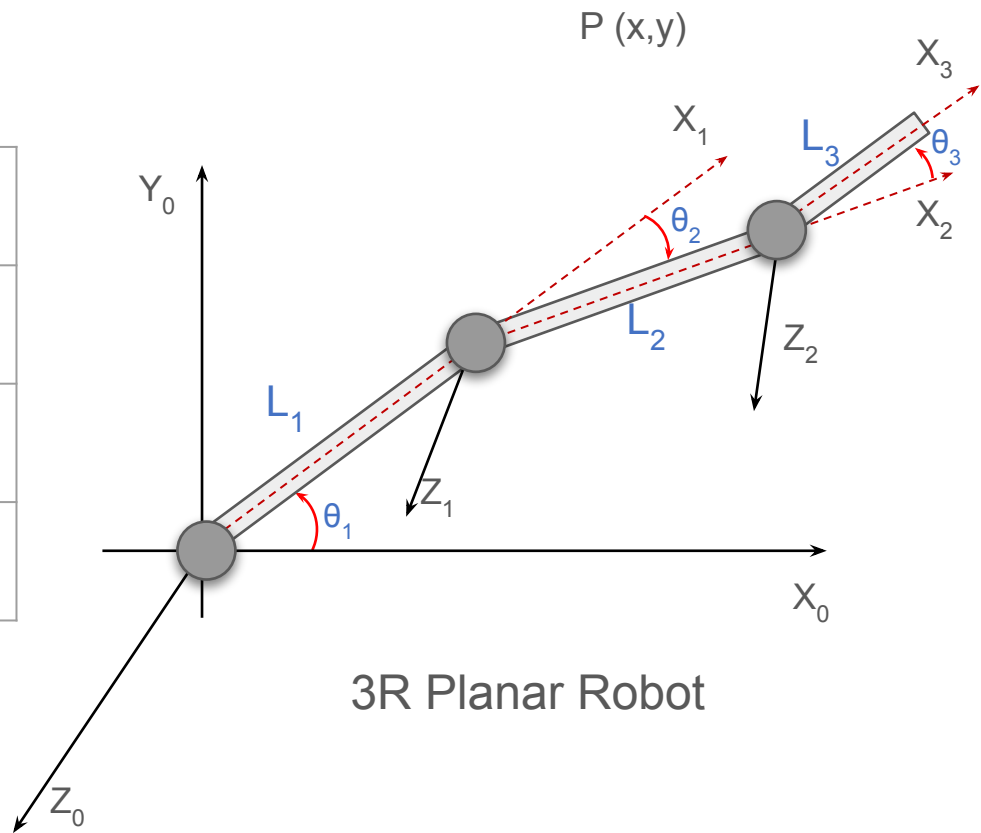
DH Table

i	θ_i	a_i	d_i	a_i
1	$+\theta_1$			
2	$-\theta_2$			
3	$+\theta_3$			



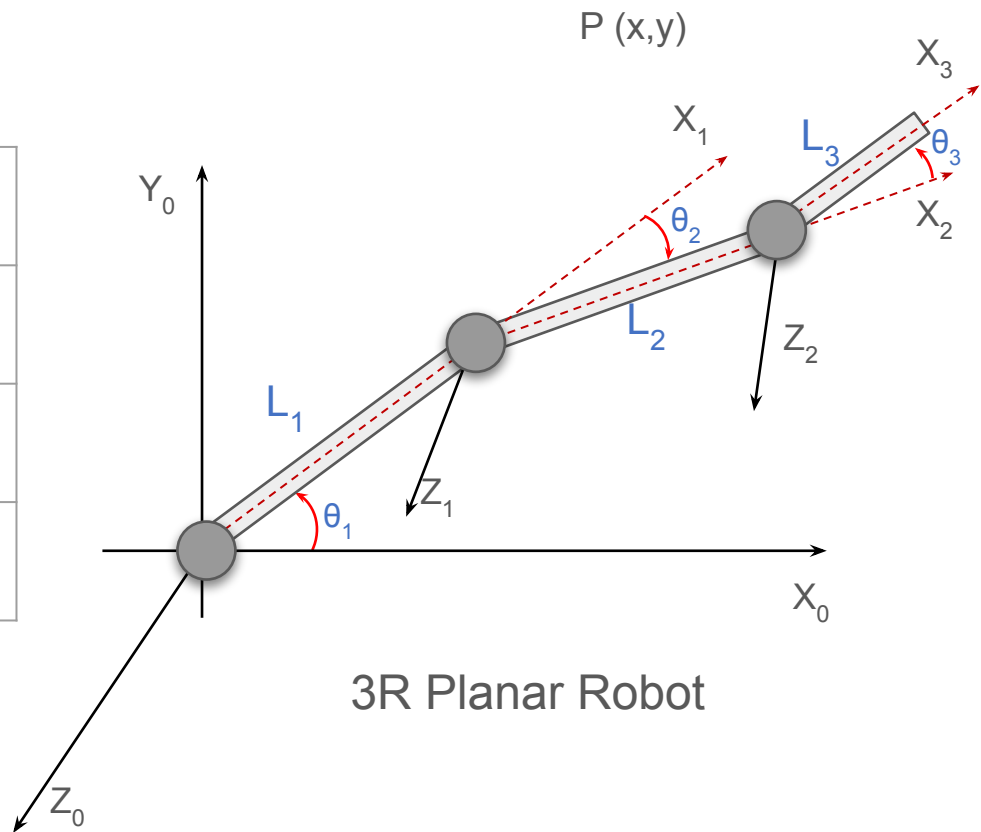
DH Table

i	θ_i	a_i	d_i	a_i
1	$+\theta_1$	L_1		
2	$-\theta_2$	L_2		
3	$+\theta_3$	L_3		



DH Table

i	θ_i	a_i	d_i	a_i
1	$+\theta_1$	L_1	0	
2	$-\theta_2$	L_2	0	
3	$+\theta_3$	L_3	0	



DH Table

i	θ_i	a_i	d_i	a_i
1	$+\theta_1$	L_1	0	0
2	$-\theta_2$	L_2	0	0
3	$+\theta_3$	L_3	0	0

